

Data Horror Escape Room: facilitator's guide and cheat sheet*

*WARNING: contains answers!!!

The Data Horror Escape Room is an online game primarily designed to be played by university staff (both researchers and research support staff) and students. It introduces and enables discussion around research data management (RDM) concepts in a fun way. It was created for an event that took place on Halloween during Data Horror Week, October 2020, so it has a spooky, Halloween theme, but the game can be played at any time of the year.

The game was designed to be played in small groups, with a facilitator supporting and helping the groups when needed, and then stimulating reflection and discussion between the groups. The facilitator should ideally, therefore, have some knowledge about RDM and be able to answer questions. The following notes are intended to help you prepare to host and facilitate an interactive session using the Data Horror Escape Room.

The Data Horror Escape Room is an open website and can be found at https://sites.google.com/vu.nl/datahorror/home.
An online escape room is a set of web pages containing puzzles that must be solved in order to 'unlock' a door that allows the participants to 'escape' and therefore complete the game. No prior knowledge of escape rooms or research data management is required to play the game.

This guide contains the following sections:

Preparation
Starting the session
Discussion points
How to play the Escape Room (⚠️ contains small spoilers)
Answers, and non-answers (⚠️ ⚠️ contains MAJOR spoilers!)
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Preparation

We highly recommend that you complete the game yourself before facilitating a session for others, and of course, you might like to recruit some friends or colleagues to play with you and help out as it is more fun to play the game with others.

To give you the same experience that your gamers will have, we recommend playing the game before reading the sections below containing spoilers (these are clearly indicated).

We advise you not to give away any information in advance that will enable your participants to find and play the game before your session. Once the game is completed, it is very easy to play again, so if people have already tried it out, it will spoil the fun.

How long does it take?

- The Data Horror Escape Room takes a *minimum* of 40 minutes with fast work and excellent team collaboration; more commonly, it takes around one hour or slightly longer.
- We recommend a session of 1 hour 30 minutes for introduction, game, and post-game discussion.
- If you have a session of less than an hour, you can start the game during the session and invite participants to complete it together, or alone, after the session. Because the room is an open website, participants can access it at any time.
- To continue the game at a later date, participants should make a note of which puzzles have been solved because the website does not track, or record any information from individual players.
- If your session isn't long enough to play the game to the end, participants will still have encountered aspects of research data management that can be discussed and you can focus on these, avoiding discussing how the puzzles are actually solved.
- Try to keep the introduction brief to maximize time for completion and allow the participants to discover for themselves how the game works. It's likely that some of your participants will have played an escape room before, and they will be able to get going very quickly. For those less experienced, offer individual support to help them on their way.

How do I set things up?

- Every team needs at least one computer and an internet connection.
- Participants will also need to make a few small notes so either you can ensure they have pens and paper, or you can suggest they make notes on their computer or mobile phone.
- At one point during the game, participants will encounter a video that they need to watch. It has sound, but can be played on mute with subtitles.
- The ideal team size is 2-6 people. You can have as many teams as you wish playing simultaneously.
- You can consider having a prize for the fastest team to complete the game.

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Notes for playing during a virtual/online event

- If playing the game during an online event, plan the following in advance:
    - how to divide the group into teams,
    - how to answer questions from the teams whilst playing the game,
    - how to send information, if needed, to all teams during the game,
    - what instructions to give teams who are not physically co-located (for example, one person will need to share their screen and control the mouse),
    - what to do when some teams have finished the game and others are still playing.
- If using Zoom, it is easiest to use the Breakout Rooms with the self-assign functionality, then participants can sort themselves into the breakout rooms of their choice. Sorting a large number of participants into breakout rooms as a facilitator is overwhelming and takes some time.

What happens when people finish?

- Consider whether, and therefore how, you would like participants to confirm they have completed the game. If they complete it during the session they might just shout out, if they complete it afterwards, you could ask them to make a screenshot when they escape, with or without timestamp.
- Make plans to keep players busy after completing the game whilst they wait for the other teams to finish. For example, you could provide prompts for each team to discuss among themselves topics raised in the game, or you could provide other RDM materials to look at and discuss.
- When all teams have finished the game, or you decide that the time is over (and they can complete the game after the session), take a look at the discussion points below.

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Starting the session

Some participants might not have previously encountered an escape room game. If so, a small amount of explanation is useful. Sometimes, other participants are keen to explain their own experiences of escape rooms to the group.

A digital escape room is a series of puzzles that must be solved in order to 'unlock' a door that allows the participants to 'escape' and therefore complete the game. The game will guide you, and there are hints, but you should be able to figure out what to do and how to do it whilst playing the game. It should be clear to you when you have completed the game: you will be told when you have escaped.

- It's usual in an escape room to have little or no explanation about the kinds of puzzles that need to be solved, or how to escape. Participants have to figure that out for themselves as part of the fun. So *don't give too much away* when explaining how an escape room works
- The room is designed to prompt discussion both during and after play, so encourage people to talk.
- You don't need to know anything about RDM before playing the game: it is not a test. But some puzzles are harder than others! A little RDM knowledge, or experience playing escape rooms, might help you complete it faster.
- Explain to participants what they should do to signal that they have completed the game.
- Explain to participants what they should do if they need help whilst playing the game. Note that on some puzzle pages there is a link to a hint that will help complete the puzzle.

Discussion points

Regardless of the audience, the game usually generates excitement and so it is worth making some time immediately after playing to reflect, de-brief, and discuss the game together.

Possible discussion points include:

- Did you like playing the game? Why (or why not!)?
- Did anything surprise you when you played the game?
- Did you learn anything, from the game itself or from your teammates?
- Which question did you find most difficult to answer?
- Do you think the Escape Room represents common issues in data management? How do these issues vary for different research areas? What issues have you personally encountered?
- Did you have discussions with your teammates? What did you discuss?
- Did anything during the game (including your struggles or mistakes) prompt you to think about why or how we should do research data management?
- Whilst playing the game, did you see any parallels between the way you organised your team and the different roles in the research process?

Of course, as a facilitator, you may want to focus on certain discussion areas, add questions that are relevant to your participants, or leave out discussion points that are not relevant.

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⚠️ *The sections below contain spoilers* ⚠️

Do not read these sections if you want to play the game without knowing the answers.

How to play the Escape Room (⚠️ contains small spoilers)

- There are 6 code puzzles, which can be solved in any order. All pages allow players to return to the central page from which they navigate to the puzzles. Each code puzzle delivers one word upon completion. The six words must be put into the correct order to unlock the final part of the game and escape.
- The 'final' puzzle is to unlock the Professor's computer and submit the data management plan. The full computer code is obtained by completing the 6 code puzzles and working out the order of the words that makes most sense.
- The game is a static website; it does not remember which puzzles have been completed or record a team's progress.
- Hints are included at the bottom of pages where people have traditionally struggled (see the answers section below). You might need to give extra tips, or help players understand an answer.
- Players have finished the game when they see the page with the open door that tells them they have escaped. It is possible that a team might believe they have completed the game before this point, so if a prize is involved, verify that they have reached the open door.
- There is no formal record of when a team completes the game (but see the note below about the team-registration version). In a live event, a team can shout out when they have finished. Facilitators can, however, give more specific instructions to provide proof that the game was completed, and a time-stamp if that is relevant: tell participants in advance that they should take a screenshot when they escape so that they can prove that they have completed the game. If you are awarding a prize for the fastest team, the file will include a timestamp in the metadata, or you can ask them to make sure that the computer screen's clock is on the screenshot to prove the moment of escape.

NOTE on the **team-registration version** of the game
There is a duplicate version of the Escape Room that is only available when facilitated by one of the game's creators and which allows teams to identify themselves and register their completion time. In this team-registration version, players have to indicate their team name at the last moment before they escape and a timestamp is recorded for that team. The facilitators at VU have access to the environment where the team names and timestamps are collected and can then identify the winning team. If interested, reach out to rdm@vu.nl

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Answers, and non-answers (⚠️ ⚠️ contains MAJOR spoilers!)

These are the puzzles and resulting code words that are delivered in the game:

Escape Room location	RDM challenge	Solution	Code word
Address book	What constitutes personal data	1. All answers are correct. 2. D-B-E-C-A	Good
Wall map	Transferring data across borders	All answers are correct.	Planning
Notice board	Persistent identifiers	Pablo Colunga-Salas's Orcid is: 0000-0002-1355-0939. This Orcid code identifies the letters on the coded sheet: HELPS. See below for more information on this puzzle.	Helps
Filing cabinet	Metadata	1-2-4-6-8-10-12	Horror
Camera	FAIR	C, D, E, G	Data
Waste paper bin	Preserving data	All answers are correct - all answers are correct - True - True - CE - ABE - False	Avoid

Complete solution for unlocking the computer:

GOOD PLANNING HELPS AVOID DATA HORROR

Where people might get stuck..

- Looking at the posters outside the room gives no useful information for completing the game, but is fun nonetheless!
- The pile of waste paper in the bin is a distraction but might prompt discussion about how, and whether, data should be destroyed.
- The map on the wall with numbers is a distraction that does not help with solving the puzzles, it is included merely for fun.
- The notice board game is the most difficult - see the section below for more explanation. Players might find it better to come back to this puzzle after completing some of the other puzzles.
- You are asked to read the data management plan (DMP) before submitting it - but this is not necessary to complete the game. All you have to do is to click 'Submit'. During discussion, you might like to return to the DMP (or your own institution's) to discuss data management planning in more detail.

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- Opening the data management plan is not quite the end… You have to click on the Submit button, and then click the final link to see the open door and the final message: "Thank you for submitting your Data Management Plan. We are pleased to inform you that your DMP has been accepted. You are free now to leave the office".
- At the bottom of the final page, it's possible to leave feedback for the creators of the Data Horror Escape Room.

Pablo's Orcid - the notice board puzzle
The notice board game is the most difficult and involves taking several steps to complete:
(1) understanding that an Orcid is a unique number identifying a person, in this case someone called " Pablo",
(2) realising that the instruction on the postcard to look at a dataset will lead to Pablo,
(3) working out how to find this dataset from the DOI (digital object identifier) that is given (or spotting that there's a direct link on the DOI flyer),
(4) recognizing where to find Pablo's Orcid in the metadata of the dataset, and
(5) using the Orcid number to identify the right letters on the coded sheet back in the escape room.

Playing the game is an opportunity to encourage researchers to create their own Orcid, and to explain why persistent identifiers are needed when archiving material on the web.

- The two flyers on the notice board explain Orcid and DOIs and can be downloaded and printed out.
- The coded sheet can only be deciphered if you know the ORCID number for Pablo Colunga-Salas. Pablo is a real researcher and is the first author of the (real) dataset mentioned on the postcard.
- The DOI on the postcard can be googled to find the dataset, or the location can be found using the instructions on the DOI flyer, or there is even a direct URL-link from the example on the DOI flyer.
- The DOI leads directly to the dataset created by Pablo Colunga-Salas and his colleagues, and from where the Orcid for Pablo can be found.
- The dataset is called '*Unicellular endoparasites of bats*' and is archived in Zenodo, so it is possible that players could locate the dataset with a few words from the title mentioned on the postcard, the name "Pablo", and the word "Zenodo".
- The coded sheet also contains the word "PABLO" and other words that players might guess are relevant (metadata, ghosts, drives, means, etc.) but which are decoys. If players try enough of these variations in the final sentence, they may eventually find the right answer despite not having completed this puzzle.

Problems with the Data Horror Escape Room

We hope you enjoy playing the Data Horror Escape Room and find it useful to help raise awareness about research data management. However, if you encounter problems, please let us know: rdm@vu.nl

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More information about the Data Horror Escape Room

- A blog post about the Data Horror Escape Room after it was initially launched: https://www.digitalscholarshipleiden.nl/articles/learn-about-data-management-concepts-in-this-online-data-horror-escape-room
- The Escape Room was included at the 2020 Research Support Games Day hosted by Robert Gordon University in the UK: https://rgu-repository.worktribe.com/output/1005411/proceedings-of-the-2nd-research-support-games-day-rsgd-2
- A presentation about the Data Horror Escape Room from the LIBER Conference 2021: https://doi.org/10.5281/zenodo.5045311 (download the pdf "Play games with me experiences using gamification for online research skills training.pdf")
- The Data Horror Escape Room won a prize at iPres 2022: https://www.dpconline.org/news/celebrating-outstanding-contributions-to-ipres-2022

Inspiration and other games

Create your own escape room using Google Forms: https://www.bespokeclassroom.com/blog/2019/10/4/how-to-build-a-digital-escape-room-using-google-forms

There is a list of Research Support games (including the Data Horror Escape Room) at: https://doi.org/10.6084/m9.figshare.16652701.v5. This includes games about all sorts of topics within the larger umbrella of Research Support and is not restricted to RDM games.

Our GHOST collective created more Escape Rooms which you can find here https://vu.nl/en/stories/making-research-data-management-fun
We also have a card game about Open Science https://zenodo.org/records/10058056

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If you need to use parts of this cheat sheet whilst facilitating a session using the Data Horror Escape Room, we have given it a Creative Commons open licence, meaning you can do so without needing to credit us. See https://creativecommons.org/publicdomain/zero/1.0/

We hope you enjoy playing and using the game as much as we have.